

# Aman Shankar Singh

📧 singhAman05 ✉ amanshsingh371@gmail.com ☎ +91-9599742303 in amanshankarsingh

## Experience

### Associate Software Engineer

Blue Yonder

Bengaluru, KA

July 2025 – Present

- Designed and automated provisioning of **customer-isolated Redis clusters** in Azure using Azure serverless architecture and CI/CD pipelines, reducing **read latency by 78%** and enabling **scalable multi-tenant deployments** with strong isolation boundaries between customer environments.
- Architected automated creation of **Azure Queues and Dead-Letter Queues (DLQ)** to improve fault tolerance and retry handling, decreasing message loss by **63% under high-latency conditions**.
- Improved system reliability and observability by implementing **structured logging and monitoring** integrations across Azure services; leveraged containerized deployments with **Docker** to enforce resource and security isolation per service.
- Developed an **MCP (Model Context Protocol) server** leveraged as a tool by agentic AI workflows to autonomously trigger customer environment provisioning, reducing environment setup **wait time** and eliminating manual dependency on multiple team members.

### Software Engineer Intern

Blue Yonder

Bengaluru, KA

Jan 2025 – June 2025

- Built a **GitHub Actions-based CI/CD workflow** and Azure DevOps to automate lifecycle management of Kafka topics, namespaces, and subscriptions on Linux hosts, improving deployment consistency across environments.
- Developed a configurable deployment interface for synchronized product solutions, reducing manual configuration overhead and improving release reliability.

## Projects

### Linux Container Runtime (Rust)

[Github Repo](#) 

- Built a **minimal OCI-compliant container runtime in Rust** that uses Linux **namespaces** (PID, NET, MNT, UTS, IPC) and **cgroups v2** to create isolated, resource-constrained process environments – mirroring the isolation model used by microVM and serverless platforms.
- Implemented **seccomp-bpf syscall filtering** to restrict the attack surface of containerized workloads, applying an allowlist policy that blocks dangerous syscalls while permitting standard POSIX operations.
- Leveraged **Linux KVM APIs** (via the `kvm` crate) to prototype guest vCPU creation and memory slot management, gaining hands-on experience with the hypervisor interface underpinning Firecracker microVMs.
- Designed a **REST API control plane** (using Actix-web) for lifecycle management (create, start, stop, destroy) of container instances, with structured logging via `tracing` crate for observability.

### Finance Tracker

[Github Repo](#) 

- Built a full-stack personal finance platform using **Next.js, React, Redux Toolkit** on the frontend and **Express.js with TypeScript** on the backend, with transactions, budgets, and bills designed as **atomically consistent, inter-linked entities**.
- Enforced **referential integrity** across the financial domain using **Supabase (PostgreSQL)**, ensuring cascading deletes and updates maintain a coherent financial state at all times.
- Leveraged **Redis** to cache derived aggregates invalidated on any write; implemented **NextAuth with Google OAuth** and rate-limiting middleware for API protection.

## Education

### Motilal Nehru National Institute of Technology, Allahabad

B.Tech in Electrical Engineering

Dec 2021 – May 2025

- CGPA: 7.86/10.00
- Coursework:** Power Systems, DSOS, Cyber Physical Systems, Microprocessor

### Ch. Chabbil Das Public School

Class-12

Mar 2019 – April 2020

- Percentage: 95.2%
- Coursework:** Physics, Chemistry, Mathematics, English, Computer Science

## Technical Skills

**Languages:** Rust, C++, TypeScript, JavaScript, Python, Java, SQL

**Systems & Linux:** Linux Namespaces, cgroups v2, KVM, seccomp-bpf, POSIX threads, epoll, Docker

**Backend & APIs:** Node.js, Express.js, Actix-web (Rust), Django, RESTful APIs

**Cloud & DevOps:** Azure, AWS, CI/CD (GitHub Actions), Azure App Services, Azure Functions, Kafka

**Databases & Caching:** MongoDB, Postgres, MySQL, Redis, Firebase

**Monitoring:** Grafana, Umami